

CoRoT-2b

R-0203

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_180905 · created 2026-07-16T22:42:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458366.6991 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.062
Transit depth (Rp/Rs)^2	0.00019 +/- 0.017 [%]
Orbital Inclination (inc)	85.54 +/- 2.02
Airmass coefficient 1 (a1)	1.32 +/- 0.16
Airmass coefficient 2 (a2)	-0.23 +/- 0.25
Scatter in the residuals of the lightcurve fit is	11.81 %
Best Comparison Star	#2 - [192, 392]
Optimal Aperture	3.4
Optimal Annulus	4.24
Transit Duration (day)	0.071 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.062 vs 0.1667 (deviation 2.56σ)
Duration fit vs published	0.071 d vs 0.095 d (deviation 1.64σ)
Mid-time O–C	3.9 min
Depth SNR	0.0
Residual scatter	11.81 %

Light curve

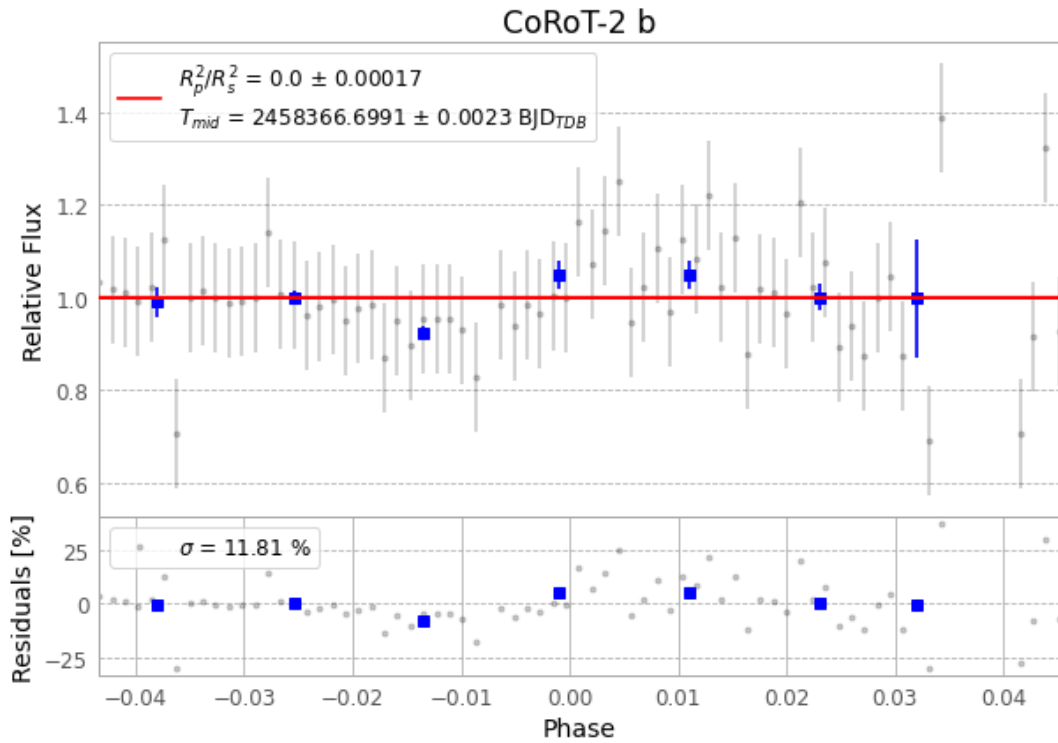


Figure 1 — FinalLightCurve_CoRoT-2 b_2018-09-04.png (SHA-256 acf6cb31bddefa94...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 251], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 38636be2d153f28a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

77 FITS frames (51 MB), CoRoT-2180905025112.FITS → CoRoT-2180905063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180905.log (3f9c15a8fbb5...), FinalLightCurve_CoRoT-2 b_2018-09-04.png (acf6cb31bdde...), FinalLightCurve_CoRoT-2 b_2018-09-04.csv (80a5d5998f16...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 22:42Z.